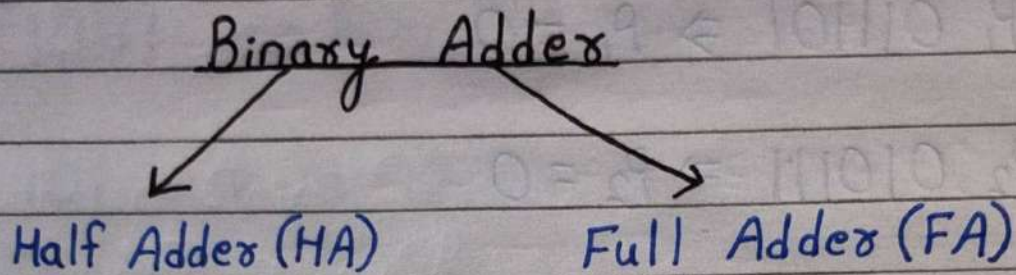


Module - II

Combinational Logic

Combinational Logic Circuit



Half Adder

Truth Table:

	A	B	Sum (S)	Carry (C)
0	0	0	0	0
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

K-map:

A \ B	$\bar{B}$	B
$\bar{A}$		1
A	1	

A \ B	$\bar{B}$	B
$\bar{A}$	0	1
A	2	3

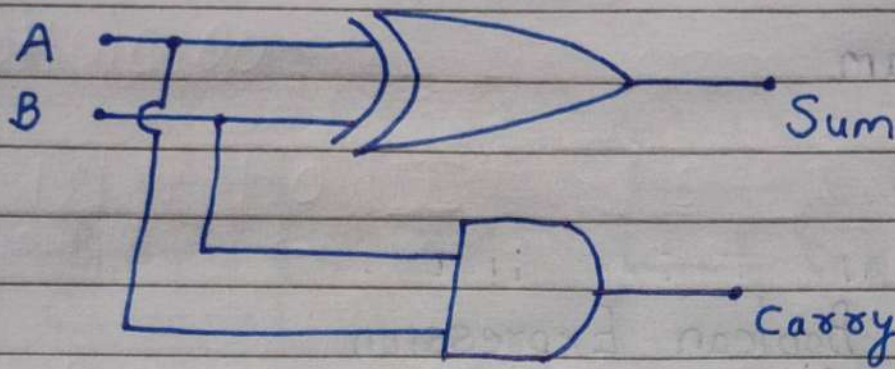
$$\text{Sum} = \bar{A}B + A\bar{B}$$

$$\text{Carry} = AB$$

$$\Rightarrow \text{Sum} = A \oplus B \text{ [XOR]}$$

$$\Rightarrow \text{Carry} = A \cdot B \text{ [AND]}$$

Logic Circuit:

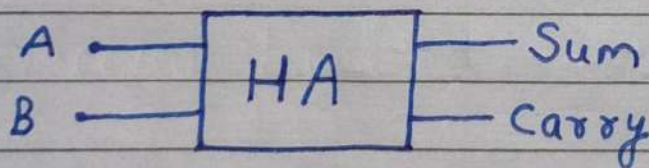


Example:

$$\begin{array}{rcccccc} & 1 & 1 & 1 & 1 & 0 & \\ & 1 & 1 & 0 & 1 & 1 & 1 \quad \leftarrow A \\ + & 1 & 0 & 1 & 1 & 0 & 0 \quad \leftarrow B \\ \hline 1 & 1 & 0 & 0 & 0 & 1 & \end{array}$$

Here, if a carry is obtain then another half adder is required, which is the limitation.

Block Diagram:

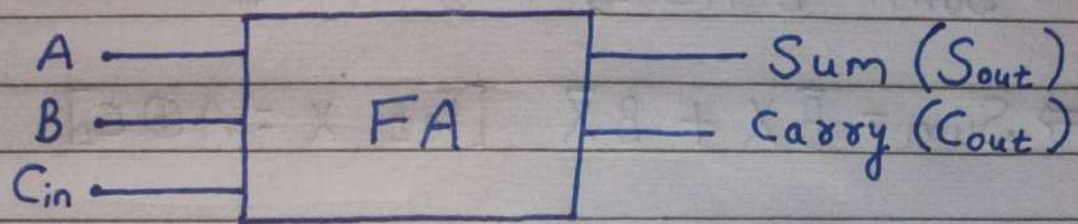


Design of Any Logic Circuit [Steps]

1. Block diagram
2. Truth table
3. K-map for Boolean Expression
4. Design
 - Basic Gate
 - NAND or NOR
 - Multiplexer

Full Adder

Block Diagram:



Truth Table:

	A	B	C _{in}	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

K-map:

A \ B C _{in}	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC_{in}	$B\bar{C}_{in}$	
$\bar{A}$		1		1	← For Sum
A	1		1		

$$\text{Sum} = A\bar{B}\bar{C}_{in} + \bar{A}\bar{B}C_{in} + ABC_{in} + \bar{A}B\bar{C}_{in}$$

$$\Rightarrow \text{Sum} = \bar{B}(A\bar{C}_{in} + \bar{A}C_{in}) + B(AC_{in} + \bar{A}\bar{C}_{in})$$

$$\Rightarrow \text{Sum} = \bar{B}(\underset{\substack{\uparrow \\ \text{XOR}}}{A \oplus C_{in}}) + B(\underset{\substack{\uparrow \\ \text{X-NOR}}}{A \odot C_{in}})$$

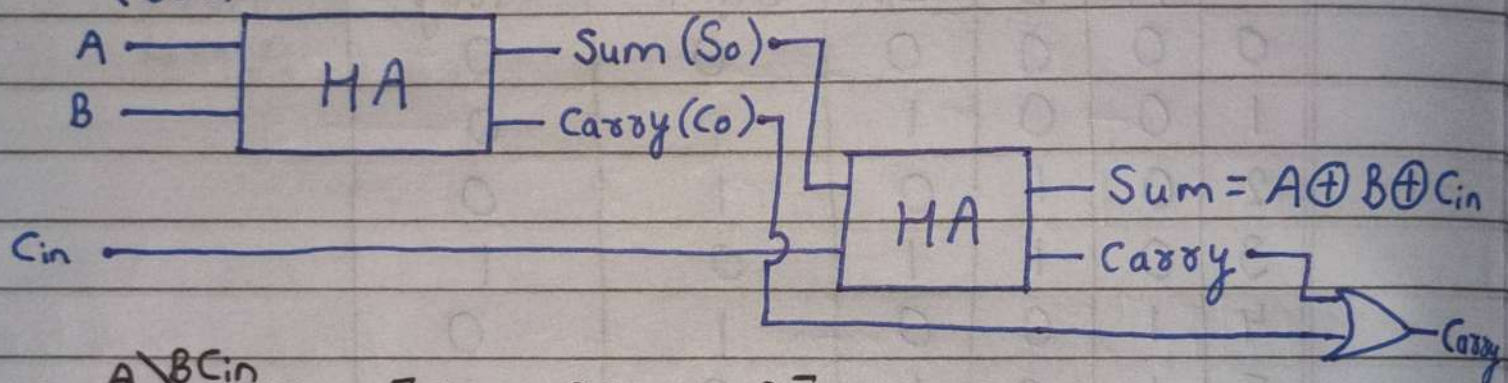
$$\Rightarrow \text{Sum} = \bar{B}(A \oplus C_{in}) + B(\overline{A \oplus C_{in}})$$

$$\Rightarrow \text{Sum} = \bar{B}X + B\bar{X} \quad [\text{let } X = A \oplus C_{in}]$$

$$\Rightarrow \text{Sum} = B \oplus X$$

$$\Rightarrow \text{Sum} = A \oplus B \oplus C_{in}$$

(Sout)



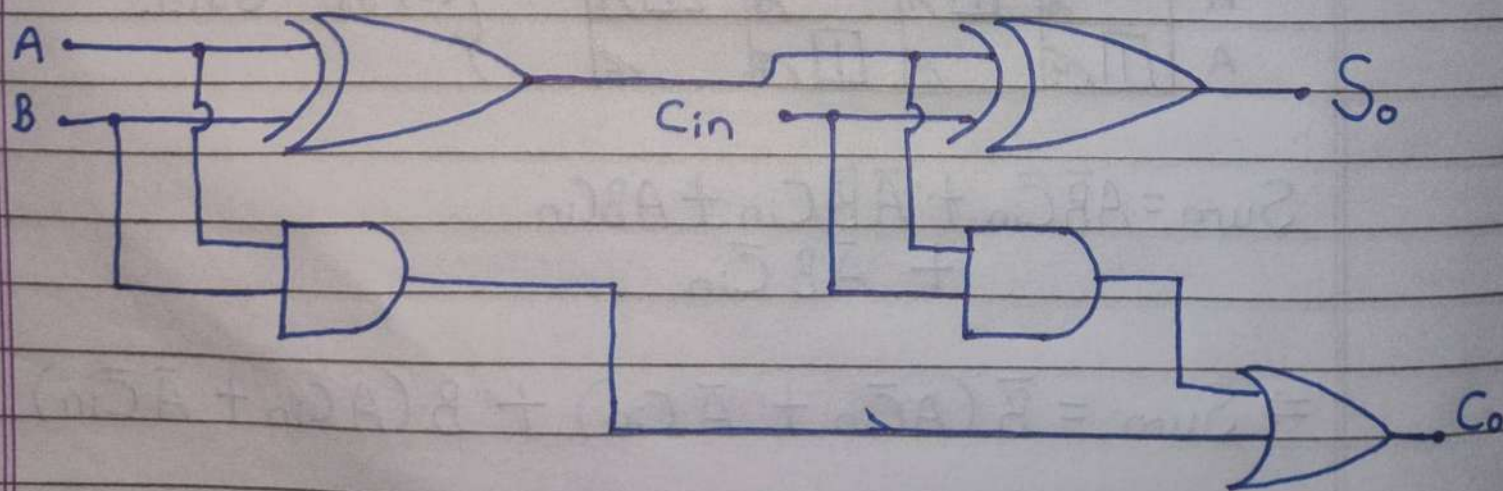
A \ BC _{in}	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC_{in}	$B\bar{C}_{in}$
$\bar{A}$	0	1	1	2
A	4	1	1	1

← For Carry

$$\text{Carry} = BC_{in} + AC_{in} + AB$$

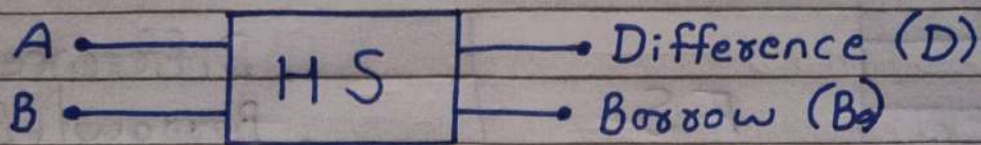
(Cout)

Logic Circuit:



Half Subtractor

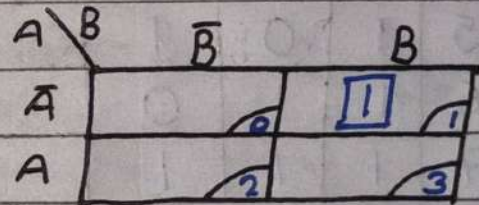
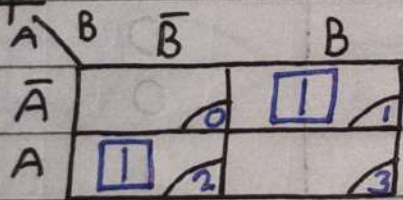
Block Diagram:



Truth Table:

	A	B	Difference (D)	Borrow (B₂)
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

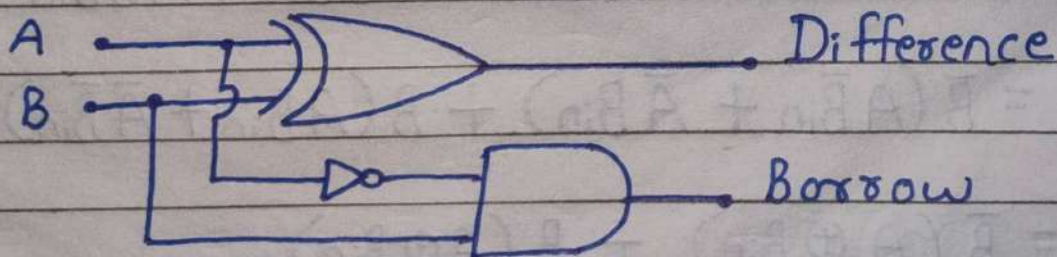
K-map:



$$\begin{aligned}\text{Difference} &= A\bar{B} + \bar{A}B \\ &= A \oplus B \\ &\quad [\text{XOR}]\end{aligned}$$

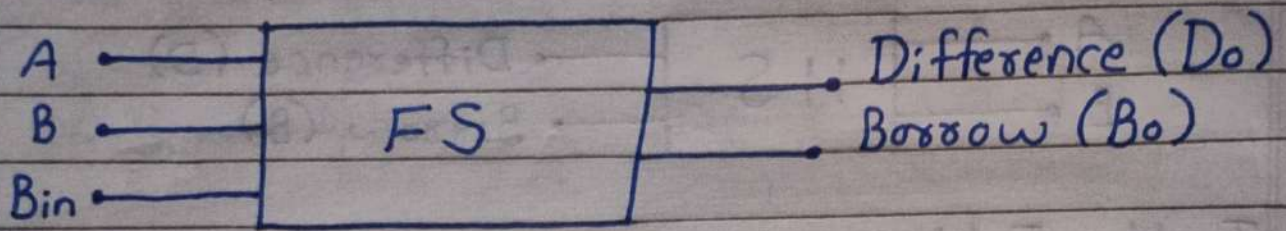
$$\text{Borrow} = \bar{A}B$$

Logic Circuit:



Full Subtractor

Block Diagram:



Truth Table:

	A	B	Bin	Difference (Do)	Borrow (Bo)
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

K-map:

	$\bar{B} \bar{B}_{in}$	$\bar{B} B_{in}$	$B \bar{B}_{in}$	$B B_{in}$
$\bar{A}$	0	1	3	2
A	4	5	7	6

← For D_o

$$\text{Difference } (D_o) = A \bar{B} \bar{B}_{in} + \bar{A} \bar{B} B_{in} + A B \bar{B}_{in} + \bar{A} B B_{in}$$

$$\Rightarrow D_o = \bar{B} (A \bar{B}_{in} + \bar{A} B_{in}) + B (A \bar{B}_{in} + \bar{A} B_{in})$$

$$\Rightarrow D_o = \bar{B} (A \oplus B_{in}) + B (A \odot B_{in})$$

$$\Rightarrow D_0 = \bar{B} (A \oplus B_{in}) + B (\overline{A \oplus B_{in}})$$

$$\Rightarrow D_0 = \bar{B} X + B \bar{X} \quad [\text{Let } X = A \oplus B_{in}]$$

$$\Rightarrow D_0 = B \oplus X$$

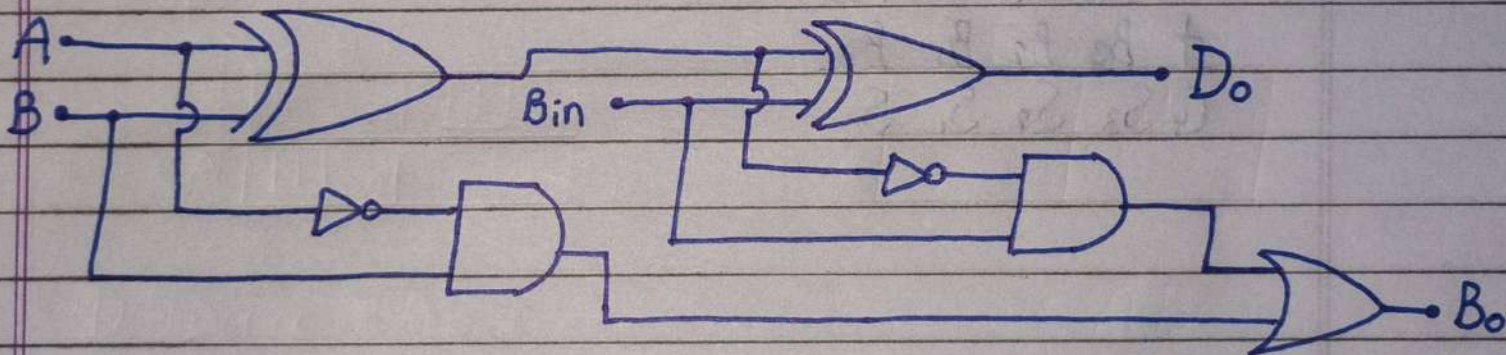
$$\Rightarrow \text{Difference } (D_0) = A \oplus B \oplus B_{in}$$

A \ B _{in}	$\bar{B} \bar{B}_{in}$	$\bar{B} B_{in}$	$B B_{in}$	$B \bar{B}_{in}$	
$\bar{A}$	0	1	1	1	} ← For B ₀
A	4	5	1	6	

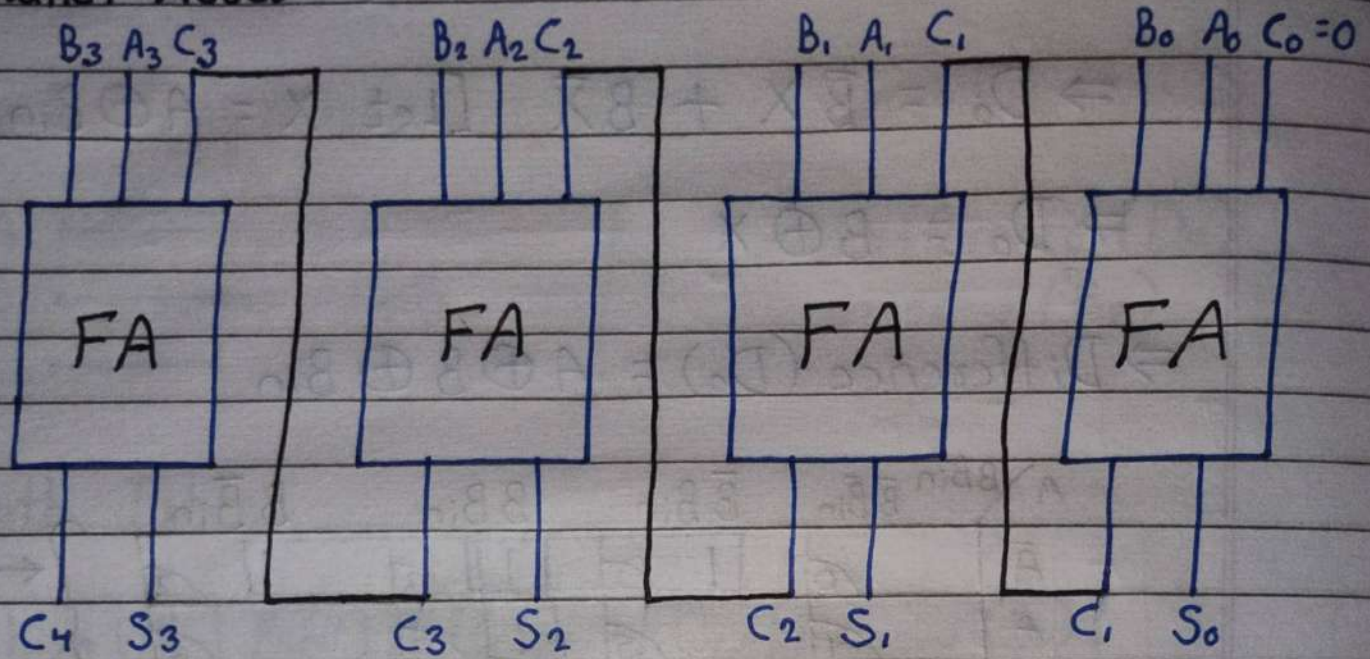
$$\text{Borrow } (B_0) = \bar{A} B_{in} + B B_{in} + \bar{A} B$$

$$\text{or } B_0 = \bar{A} B + B_{in} (\bar{A} + B)$$

Logic Circuit:



Parallel Adder



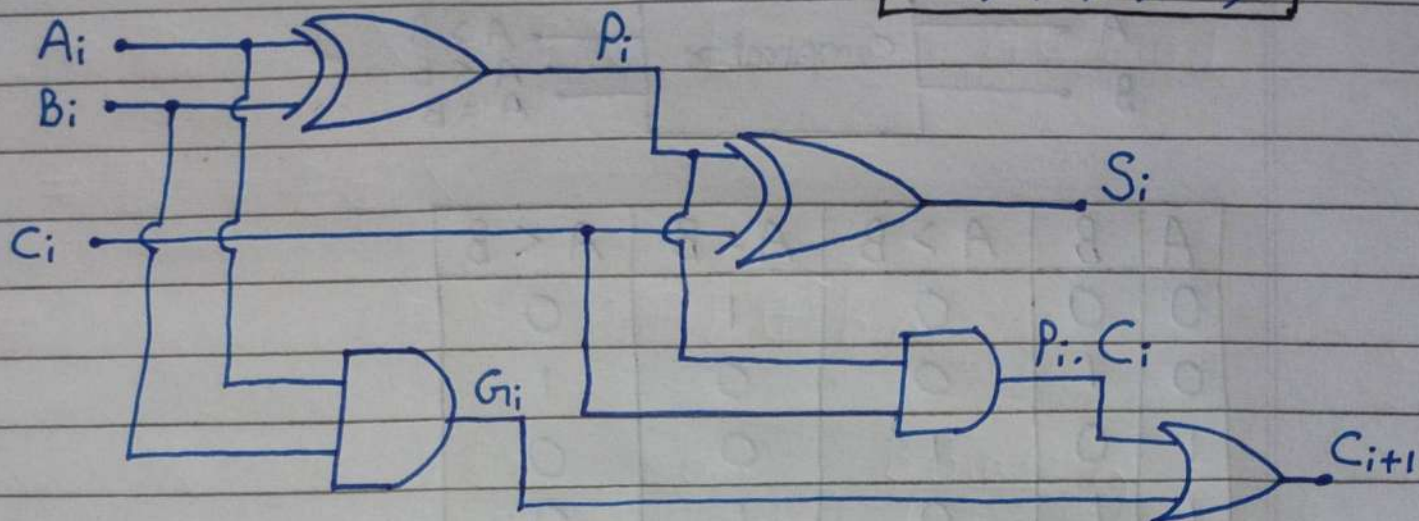
Final Sum:

C_4	S_3	S_2	S_1	S_0
-------	-------	-------	-------	-------

$$\begin{array}{r} C_4 \ C_3 \ C_2 \ C_1 \ C_0 \\ A_3 \ A_2 \ A_1 \ A_0 \\ + B_3 \ B_2 \ B_1 \ B_0 \\ \hline C_4 \ S_3 \ S_2 \ S_1 \ S_0 \end{array}$$

Look Ahead Carry Adder

$$i = 0, 1, 2, 3, \dots, n$$



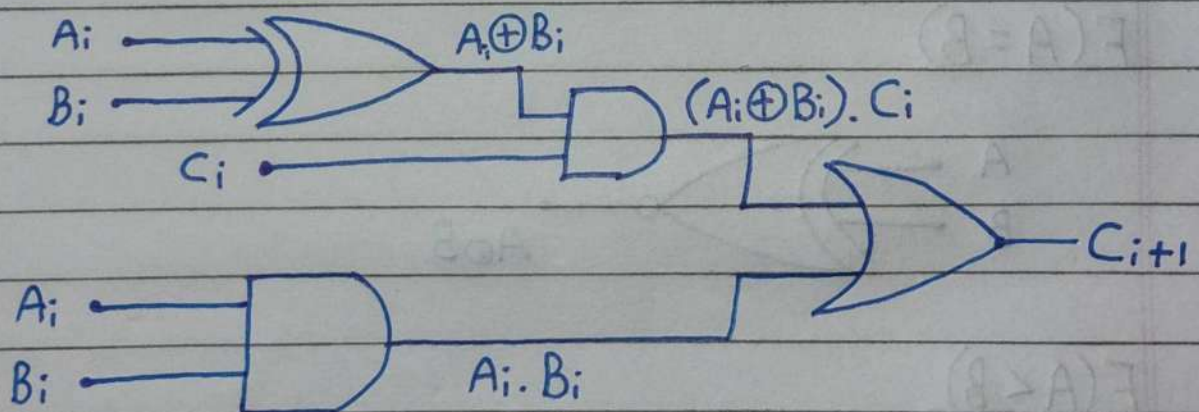
$$P_i = A_i \oplus B_i$$

$$G_i = A_i \cdot B_i$$

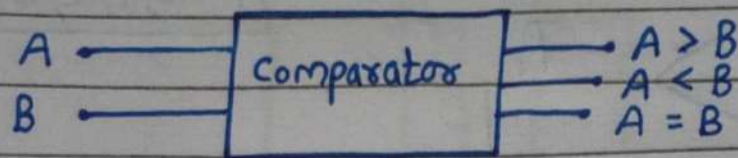
$$S_i = P_i \oplus C_i = A_i \oplus B_i \oplus C_i$$

$$C_{i+1} = G_i + P_i \cdot C_i$$

$$\Rightarrow C_{i+1} = A_i \cdot B_i + (A_i \oplus B_i) \cdot C_i$$



Comparators



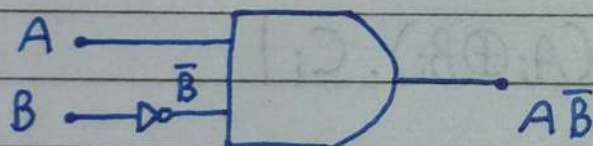
A	B	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

$$F(A > B) = A\bar{B}$$

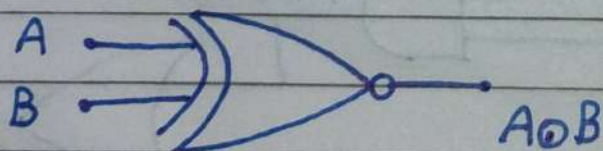
$$F(A = B) = \bar{A}\bar{B} + AB = A \odot B$$

$$F(A < B) = \bar{A}B$$

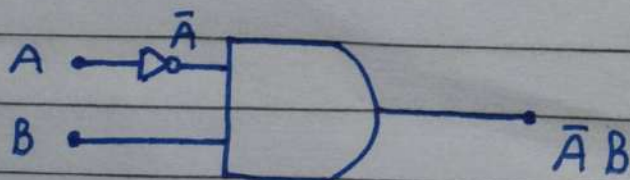
$$F(A > B)$$



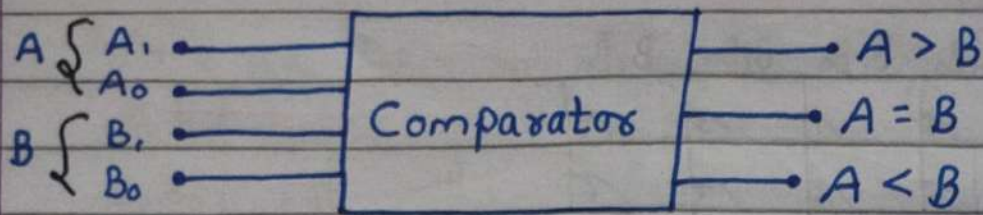
$$F(A = B)$$



$$F(A < B)$$



Comparators [2 Bit]



	A		B		A > B	A = B	A < B
	A ₁	A ₀	B ₁	B ₀			
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

K-map for $A > B$

$A, A_0 \backslash B, B_0$	$\bar{B}, \bar{B}_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	0	1	3	2
$\bar{A}, A_0$	4	5	7	6
A, A_0	12	13	15	14
$A, \bar{A}_0$	8	9	11	10

$$F(A > B) = A, \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + \cancel{A_0 \bar{B}_1 \bar{B}_0} + A, A_0 \bar{B}_0$$

K-map for $A = B$

$A, A_0 \backslash B, B_0$	$\bar{B}, \bar{B}_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	1	0	3	2
$\bar{A}, A_0$	4	5	7	6
A, A_0	12	13	15	14
$A, \bar{A}_0$	8	9	11	10

$$F(A = B) = \bar{A}, \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}, A_0 \bar{B}_1 B_0 + A, A_0 B_1 B_0 + A, \bar{A}_0 B_1 \bar{B}_0$$

$$= \bar{A}, \bar{B}_1 (\underbrace{\bar{A}_0 \bar{B}_0 + A_0 B_0}_{X-NOR}) + A, B_1 (\underbrace{A_0 B_0 + \bar{A}_0 \bar{B}_0}_{X-NOR})$$

$$= (A_0 \odot B_0) (\underbrace{\bar{A}_1 \bar{B}_1 + A_1 B_1}_{X-NOR})$$

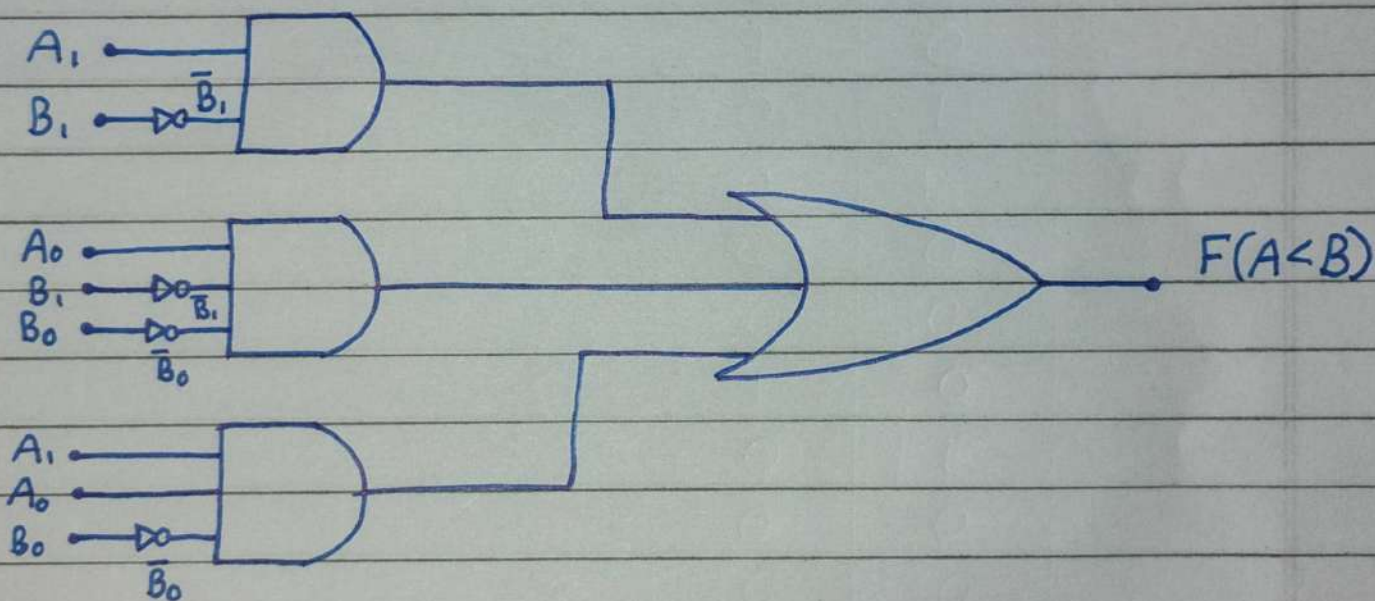
$$= (A_0 \odot B_0) (A_1 \odot B_1)$$

K-map for $A < B$

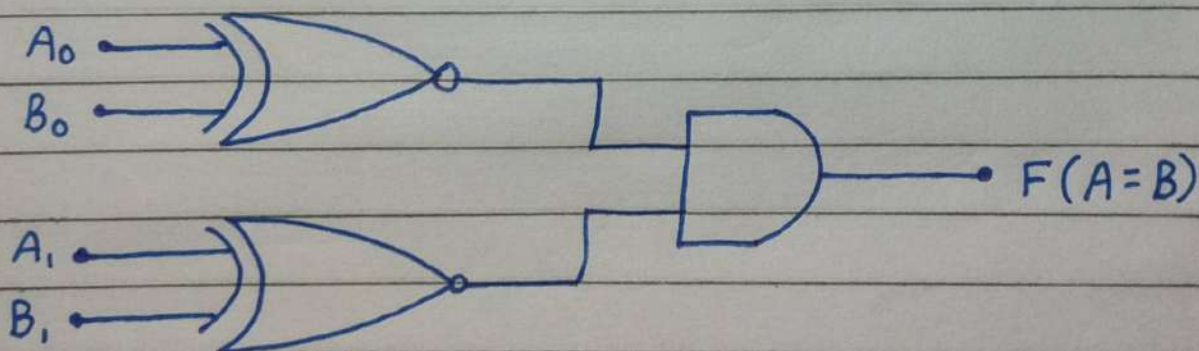
A, A_0	B, B_0	$\bar{B}, B_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	0	1	1	1	2
$\bar{A}, A_0$	4	5	1	1	6
A, A_0	12	13			14
$A, \bar{A}_0$	8	9	1	11	10

$$F(A < B) = \bar{A}_1 \bar{B}_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

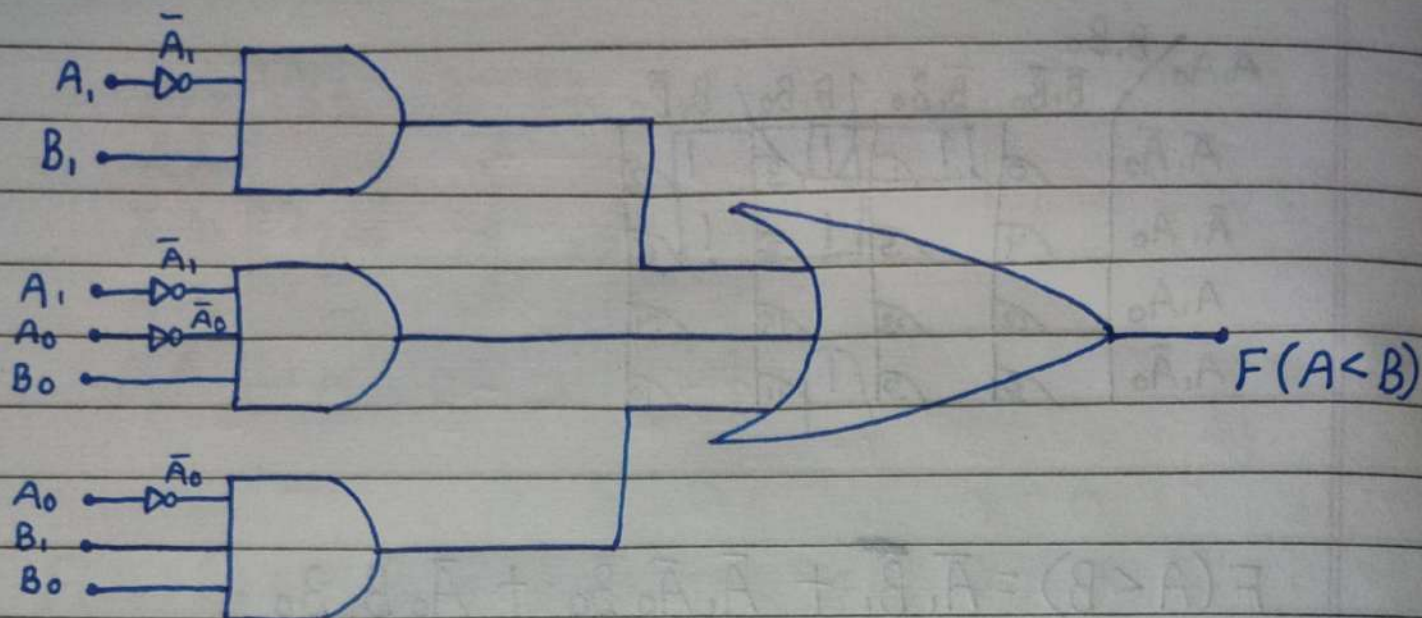
$A > B$



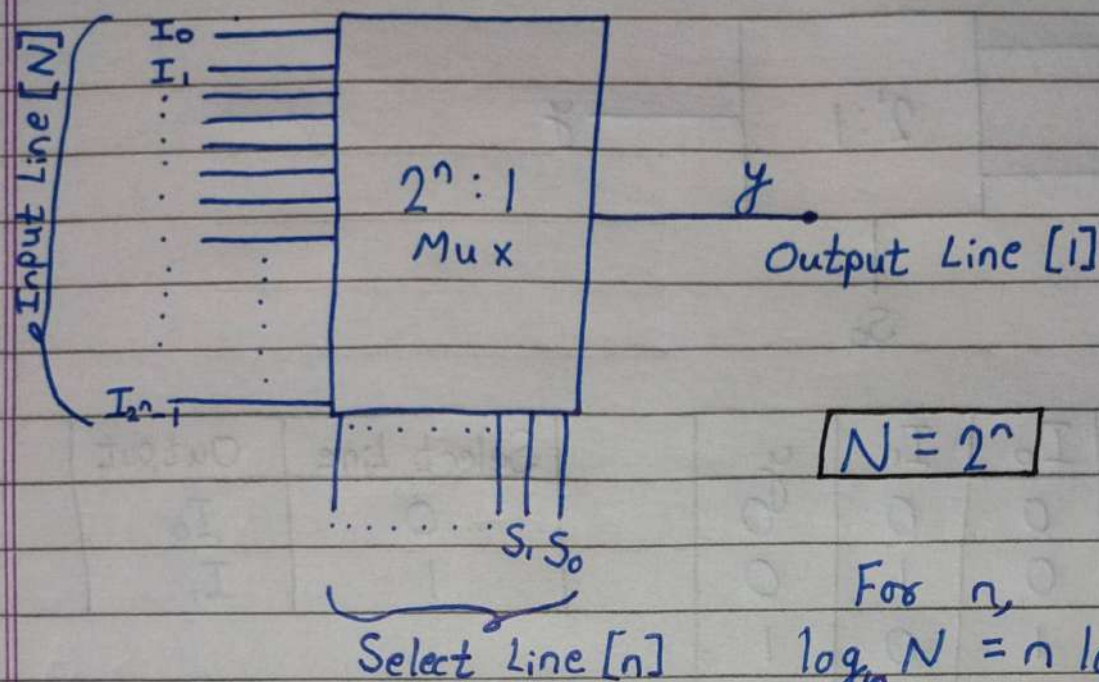
$A = B$



$A < B$



Multiplexer



$$N = 2^n$$

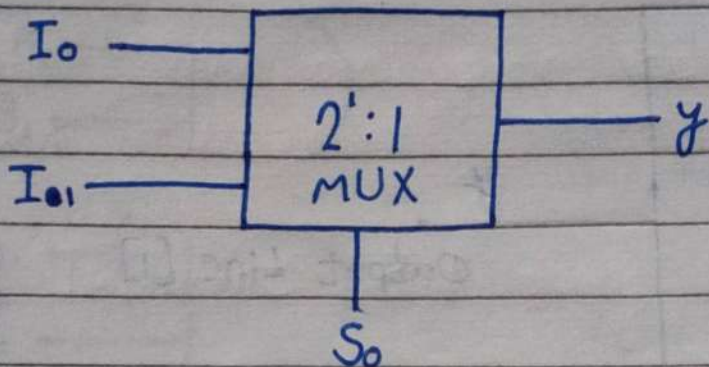
For n ,
 $\log_{10} N = n \log_{10} 2$

Uses:

$$n = \frac{\log_{10} N}{\log_{10} 2} = \log_2 N$$

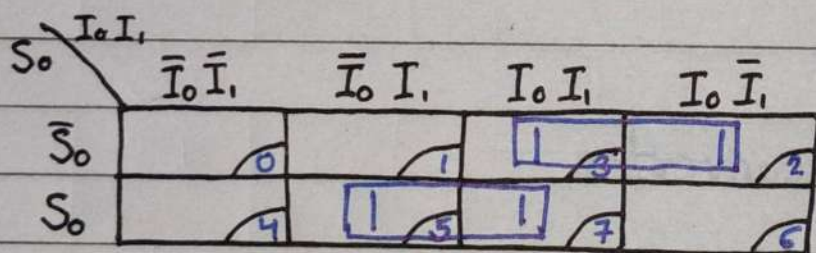
- Data selector & Data Routing
- Switch
- Many-to-one converter
- Parallel-to-serial converter
- Universal logic circuit (Implement any type of logic & gate.)

For 2:1 MUX

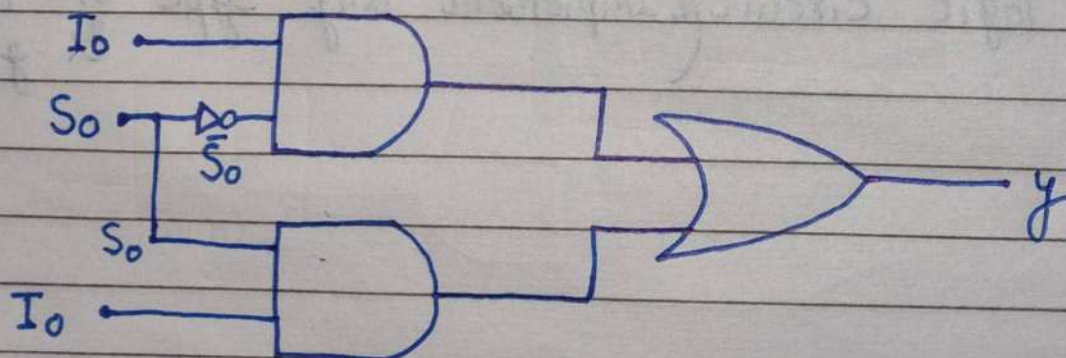


	S_0	I_0	I_1	y
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

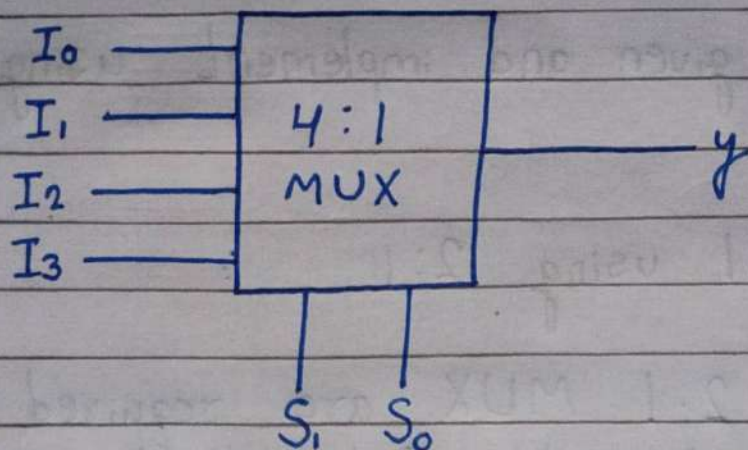
Select Line	Output
0	I_0
1	I_1



$$y = \bar{S}_0 I_0 + S_0 I_1$$

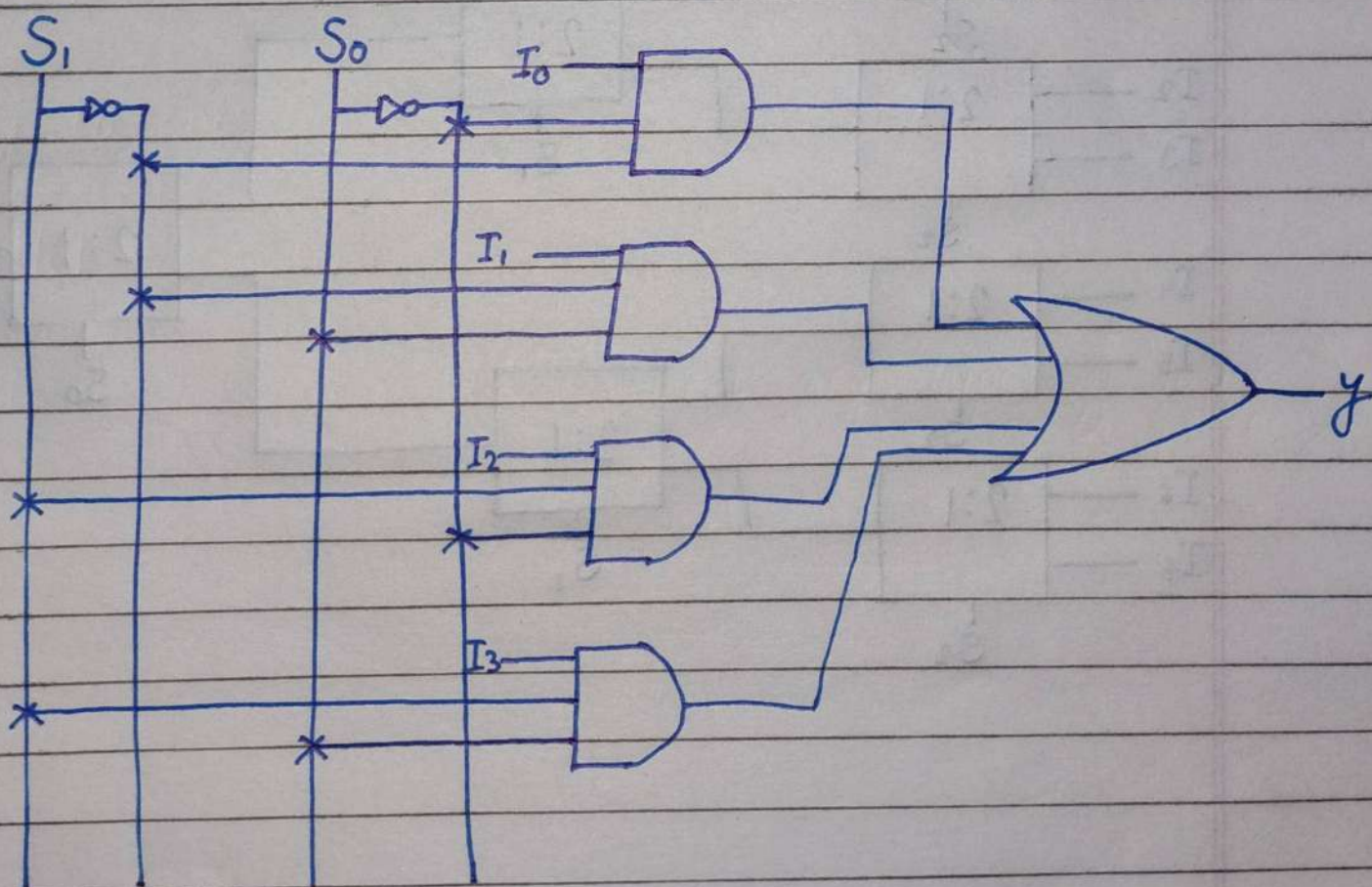


Ex 4:1 MUX



	S_1	S_0	y
0	0	0	I_0
1	0	1	I_1
2	1	0	I_2
3	1	1	I_3

$$y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$



Question Type - 1

Higher order are given and implement using lower order multiplexers.

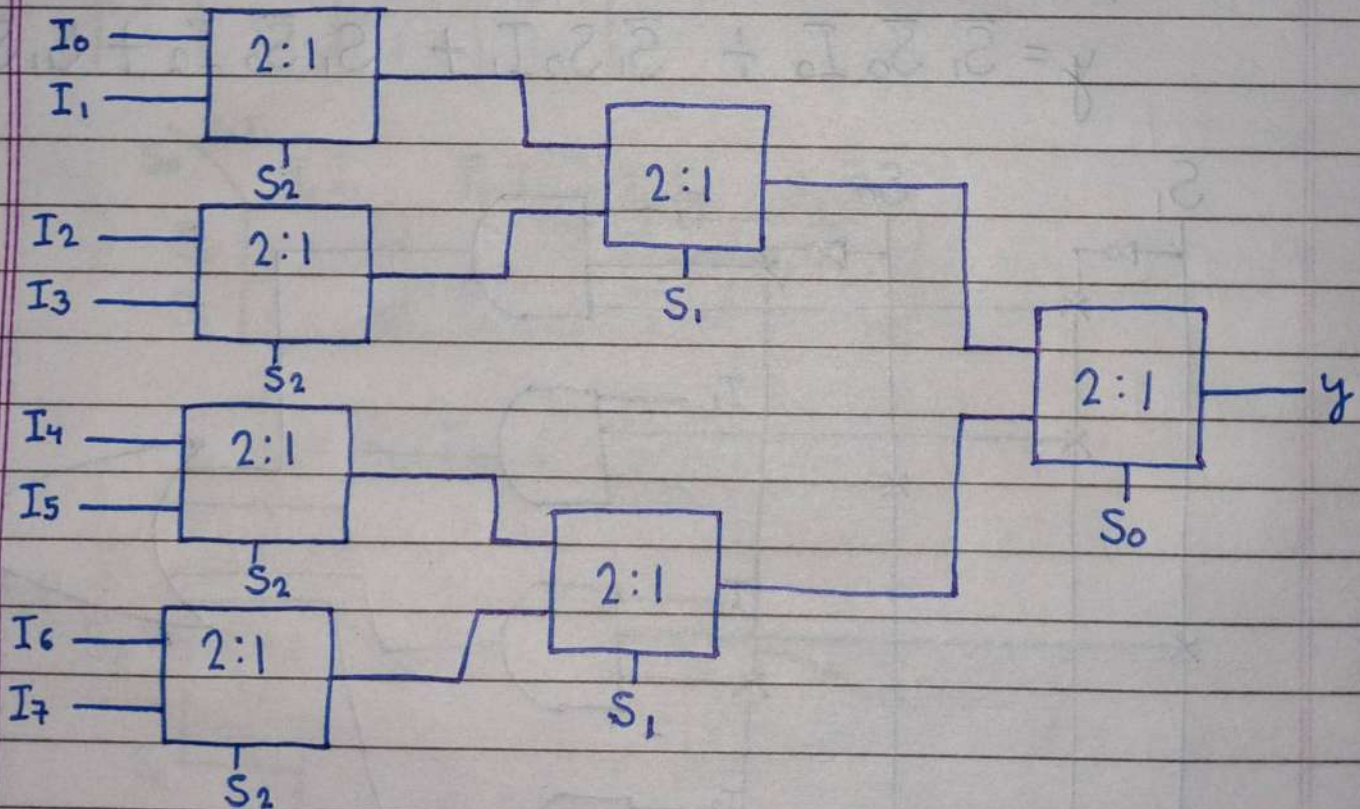
(Q) Implement 8:1 using 2:1

$$\frac{8}{2} = 4$$

$$\frac{4}{2} = 2$$

$$\frac{2}{2} = 1$$

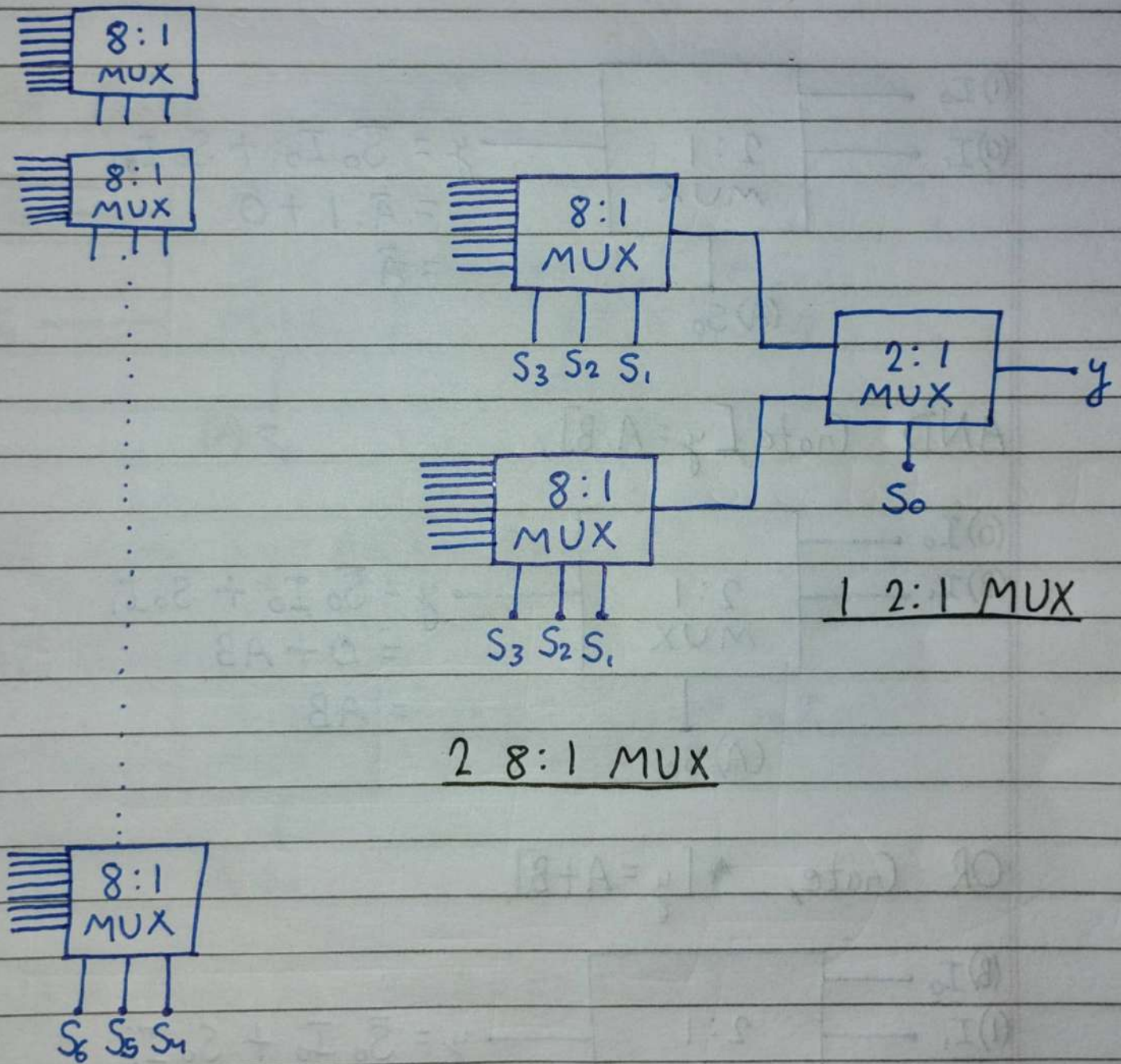
7 2:1 MUX are required to implement 8:1 MUX.



(Q) Implement 128:1 using 8:1

$$\frac{128}{8} = 16$$

$$\frac{16}{8} = 2$$

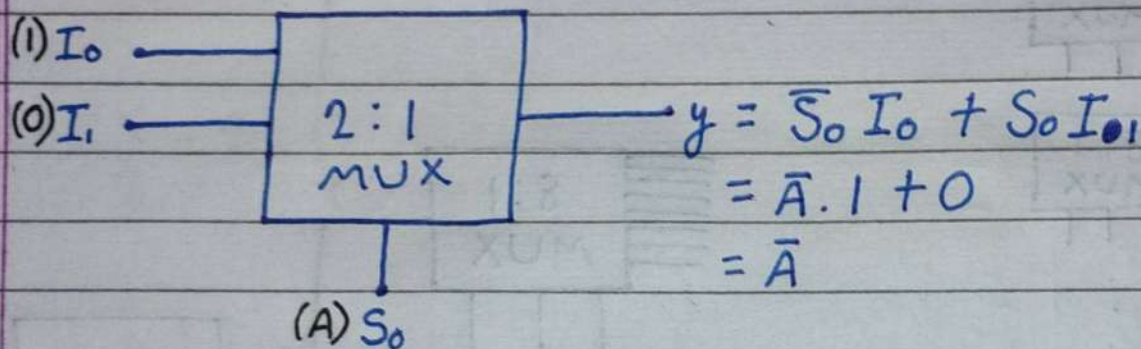
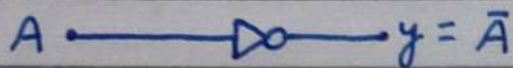


16 8:1 MUX

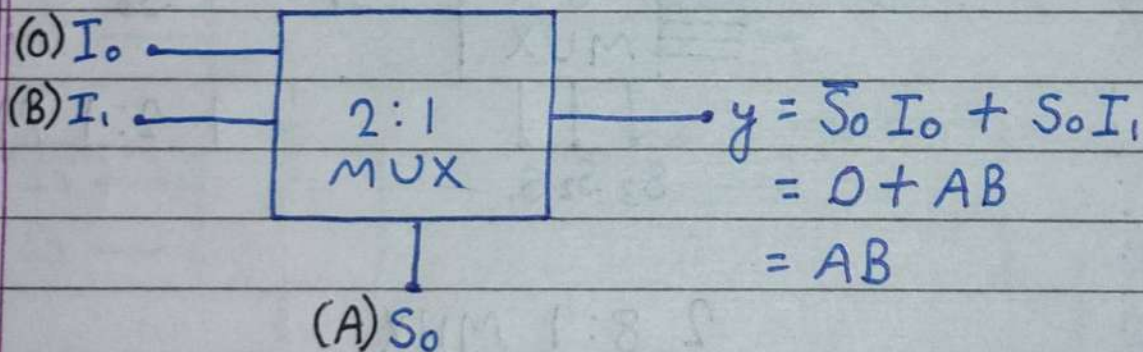
Question Type - 2

Multiplexer as a Logic Gate.

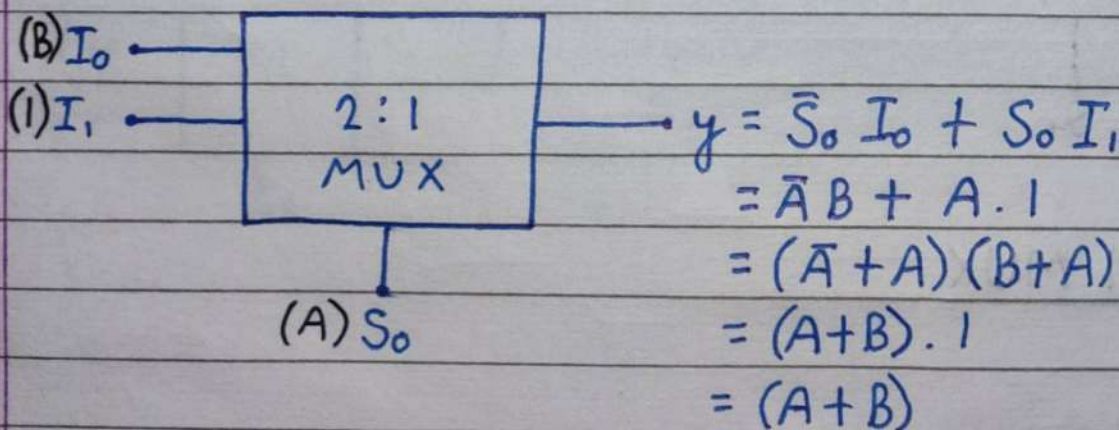
→ NOT Gate, $[y = \bar{A}]$,



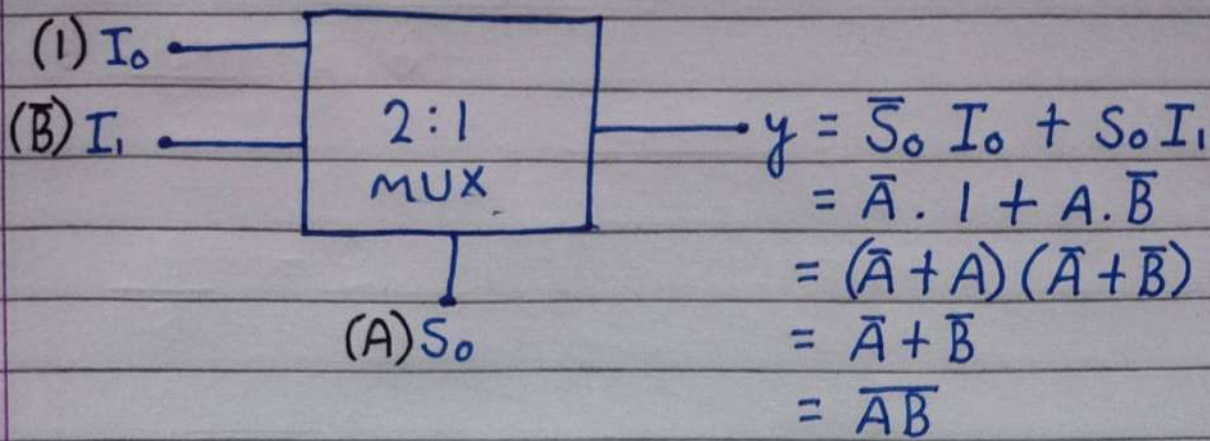
AND Gate $[y = A \cdot B]$,



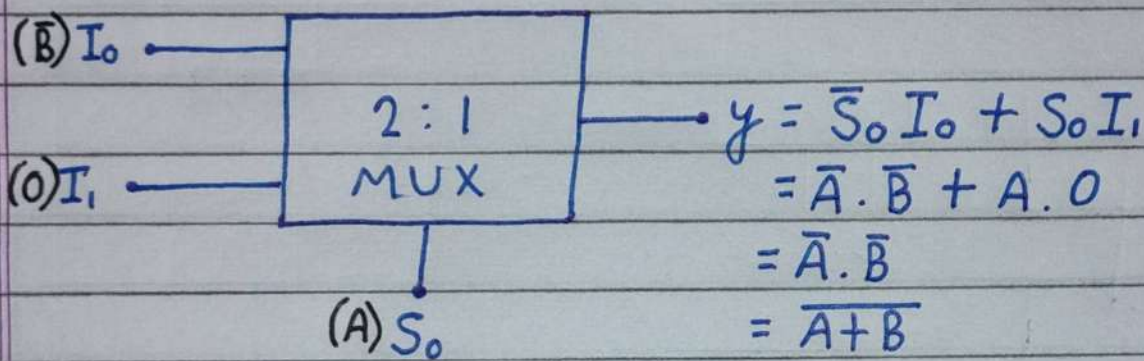
OR Gate, $[y = A + B]$,



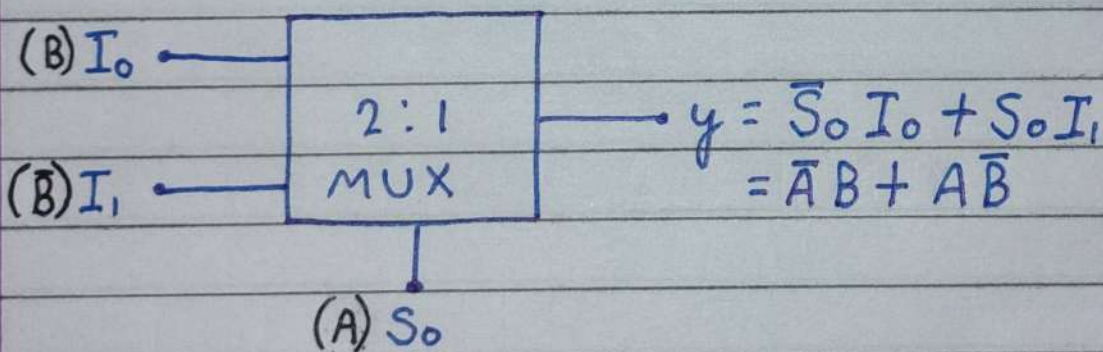
NAND Gate $[y = \overline{A \cdot B}]$,



NOR Gate $[y = \overline{A + B}]$,



XOR Gate $[y = A \oplus B = \overline{A}B + A\overline{B}]$,



X-NOR Gate $[y = A \odot B = AB + \overline{A}\overline{B}]$,

